IST 387: Introduction to Applied Data Science
Lab 1: Part 2 Notes

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Reference: Data Science for Business with R, written by Jeffrey S. Saltz and Jeffrey M. Stanton

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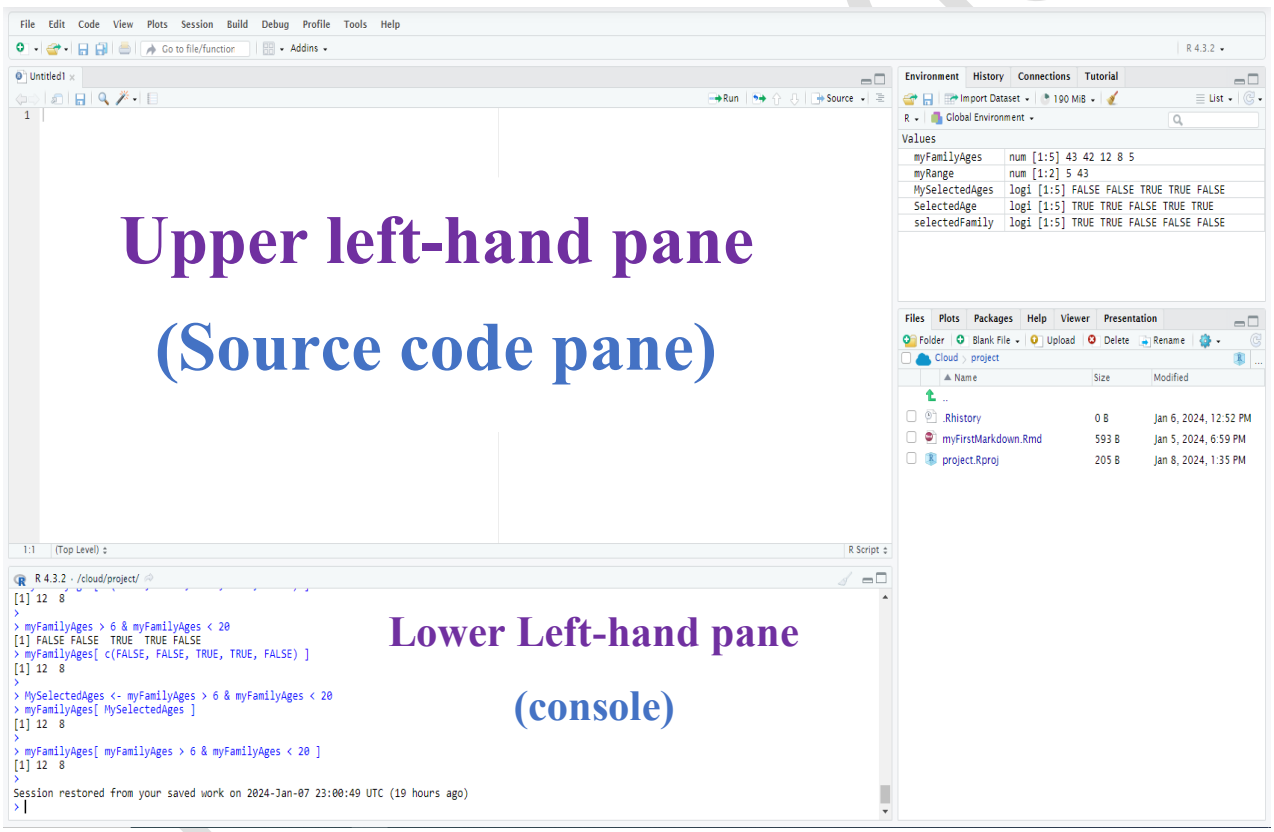
Installing RStudio

The most important aspect of installing RStudio is to install R first. Make sure that you have the latest version of R installed before you begin with the installation of RStudio. There is ample documentation om the RStudio website, http://www.rstudio.org/ , so if you follow the instructions there, you should have minimal difficulty.

If you reach a page where you are asked to choose between installing RStudio server and RStudio as a desktop application on your computer, choose the latter. If you need additional guidance about RStudio, have a look at the book titled "*Getting started with RStudio*", by John Verzani. There is also a YouTube video that introduces RStudio here: http://www.youtube.com/watch?v7sAmqkZ3Be8 .

When you run RStudio, you will see three or four sub-windows. Use the **File** menu to select **New**, and in the sub menu for **New**, select **R script**. This should give you a screen that looks as follows:



Creating R Scripts

In the figure show above, in the lower left-hand pane of RStudio, you will notice that we have a regular R console running. You can type commands into this console, just like we did earlier using plain old R. Click in the console pane and type the following:

```
> tinyData <- c(1,2,1,2,3,3,3,4,5,4,5)
> mean(tinyData)
[1] 3
```

Now let's use the upper left-hand pane, and enter the following code:

```
tinyData <- c(1,2,1,2,3,3,3,4,5,4,5)
min(tinyData)
mean(tinyData)
sum(tinyData)
```

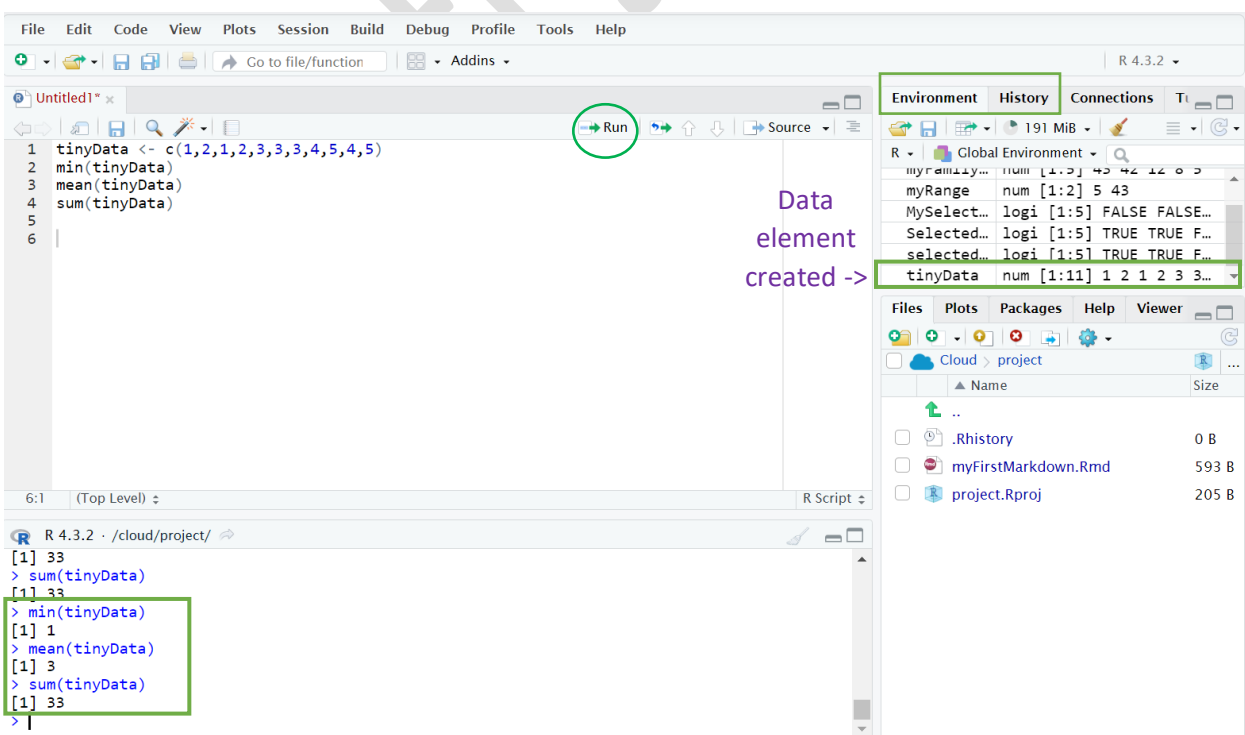
You can see, we are still writing R code, and rather than typing things at the ">" console prompt, we are just putting them in an R source code file that can be saved.

Once we have the R source code, we can click on the **Run** button to run these commands. There are two ways to use the run button:

1. Highlight the piece of code that you want to run.
2. Click in a line of code without highlighting anything and press the run button.

Once you click **Run** (refer to the figure given below)**,** R will copy your code into the **Console** and the output will appear. The advantage of this is that this code can be saved. Also, you can fix problems in the code, make copies of it, add comments, and do other things that will help you be a more productive data scientist. You should be able to see (refer to the figure below) in the upper right corner (under the environment tab), that new data element (**tinyData**) has got created.

Another tab in the upper right corner is the **History Tab.** This is useful to see the list of previous R commands we have executed.



Case Study: Calculating NPS

NPS is a measurement that many businesses use to summarize the attitudes of a set of consumers toward the product or service that the business offers. To calculate NPS, we need a vector of numbers where each number represents a *likelihood to recommend* an answer to the question: "On a scale of 1 to 10, how likely are you to recommend this [product or service]". In the present case study, the flyers have been asked how likely they were to recommend the airline that provided their flight.

Below is the code to calculate NPS (in the source code window). In this code note the use of comments (starting with the "#" character).

After writing the code and running it, you will see the output in the console window.

```
#let the test vector be ltr
ltr <- c(9,8,3,9,7,8,9,6,7,8,9)

#Calculate the length of the vector
total <- length(ltr)

#create a new vector with just the promoters
promoters <- ltr > 8
numpromoters <- ltr [promoters]

#calculate the length of the numpromoters
totalnumpromoters  <- length(numpromoters)

#create a new vector with just the detractors
detractors <- ltr < 7
numdetractors <- ltr [detractors]

#calculate the length of the numdetractors
totalnumdetractors  <- length(numdetractors)

#Calculate the NPS
nps <- (totalnumpromoters/total - totalnumdetractors/total) * 100

#output NPS
nps
```

Output in the console window

```
> nps
[1] 18.18182
```